

Low-Voltage-Driven Actuators Using Photo-Cross-Linked Ionic Columnar Liquid-Crystalline Polymer Films

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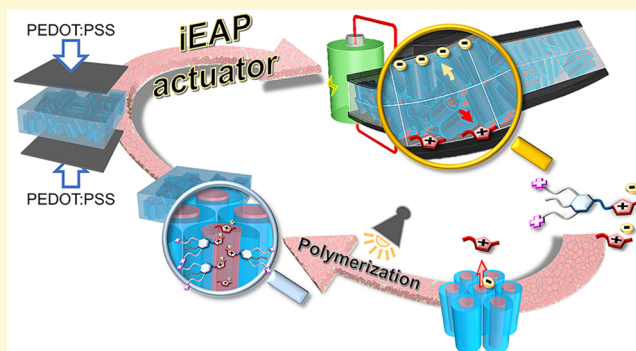


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Supporting Information

ABSTRACT: Liquid-crystalline molecular self-assembly has become an attractive strategy to enhance mass transport and mechanical strength. Herein we report low-voltage-driven actuators based on photo-cross-linked ionic columnar liquid-crystalline polymer films integrating ionic liquids into nanoscale 1D channels. We have unveiled the effect of nanochannels on the actuator performance. The nanostructured actuator exhibits larger deformation and generated force compared with the corresponding amorphous actuator. The concept of nanosegregation consisting of liquid and anisotropic rigid domains will open up a promising pathway for electroactive actuator design.



Ionic electroactive polymer (iEAP) actuators have been intensively investigated for their use in potential applications such as soft grippers, active catheters, and haptic devices.^{1–5} The large deformation induced by ion diffusion at a low driving voltage (a few volts) is beneficial for such applications compared with piezoelectric and dielectric polymer actuators that require high driving fields of several tens of $V \mu m^{-1}$. Polymer electrolytes are one of the essential components for achieving high performance in iEAP actuators. Conventionally, amorphous polymer ion gels containing ionic liquids and hydrated ionic polymers, like Nafion, have been employed as polymer electrolytes.^{6–8} However, they have drawbacks of low force generation and poor durability under long-term actuation in air, mainly because of the lack of mechanical robustness and leakage of the inner electrolytes and water. Currently, several approaches have been reported to obtain ionic-liquid-containing nanostructured polymer electrolytes with high ionic conductivities and mechanical toughness, such as using block copolymer self-assembly^{9–12} and geometric engineering in polymer blends¹³ and electrode–electrolyte interfaces.^{14,15} Although these nanostructured polymers have improved the leakage problems of electrolytes, their deformability and generation force remain insufficient. It is a great challenge to develop new nanostructured polymer electrolytes that combine the contradictory functions of high ionic conductivity and a high elastic modulus for the next generation of iEAP actuators. Our strategy is the confinement of ionic liquids into 1D ion channels formed in photo-cross-linked

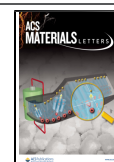
ionic columnar liquid-crystalline polymer matrices to form more mobile continuous ion transport pathways.

The self-assembly of nanosegregated liquid crystals and the subsequent interlocking of the structures by in situ photopolymerization play critical roles in producing ion-active and mechanically tough polymer membranes.^{16–21} We previously reported a diverse range of nanostructured polymer electrolytes forming 1D to 3D ion channels through the self-assembly of columnar, smectic, and bicontinuous cubic ionic liquid crystals.^{16,22,23} For example, by using in situ photopolymerization of the ionic columnar liquid-crystalline phase of a polymerizable fan-shaped imidazolium salt with diacrylate groups (DA, in Figure 1A) that formed a vertical columnar orientation on a chemically modified surface,²² we have successfully developed the first example of a one-dimensionally ion-conductive free-standing polymer film. It was found that the through-plane conductivity of the nanostructured polymer membrane was three orders of magnitude higher than the in-plane membrane conductivity at 100 °C. However, the very low conductivity at ambient temperature limited its applications for practical use in electrochemical energy devices. We

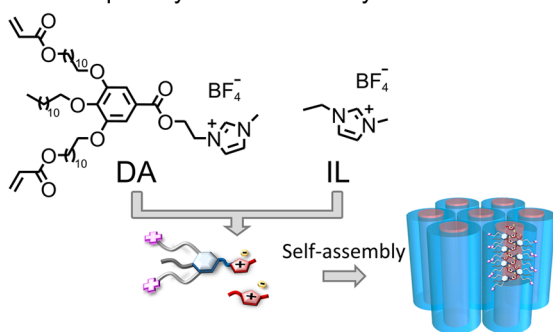
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(A) Columnar liquid-crystalline assembly



(B) Bending actuation by applying a voltage

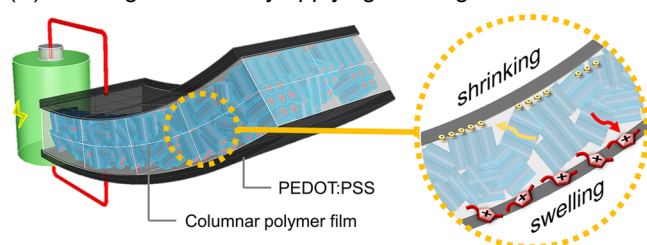


Figure 1. (A) Molecular structures of the wedge-shaped imidazolium compound with diacrylate groups (DA) and ionic liquid [EMIM][BF₄] (IL) and their self-assembly into the columnar liquid-crystalline phase forming 1D ion channels. (B) Illustration of the electric-field-driven bending actuation of the three-layer membranes composed of the columnar liquid-crystalline polymer electrolyte and PEDOT:PSS electrodes.

envisaged that the incorporation of fluidic ionic liquids into nanochannels should increase the conductivity while maintaining the continuity of the channels, which can expand their applicability in a variety of fields including iEAP actuators, carbon dioxide gas separation,²⁴ and water treatments.^{25,26}

Herein we report two voltage-driven nanostructured iEAP actuators using the ionic columnar liquid-crystalline polymer film of DA, featuring a confinement of ionic liquid (IL) into the 1D ion channels for the first time (Figure 1). The free-standing flexible nanostructured membrane organizing 30 mol % of IL (7.9 wt %) was sandwiched between two conductive polymer electrodes, poly(3,4-ethylenedioxythiophene) doped with poly(styrenesulfonate) (PEDOT:PSS). We successfully demonstrated the faster and larger bending displacement and higher force generation for the nanostructured polymer membrane actuators compared with the corresponding non-ordered actuator using the amorphous membrane obtained by the photopreservation of DA in the isotropic liquid (Iso) phase.

Selecting the appropriate ionic liquids as nonvolatile lyotropic solvents and electrolytes is the key to inducing nanostructured liquid-crystalline phases and achieving high ionic conductivity. To obtain room-temperature columnar liquid crystals confining ionic liquids into the ionic channels of DA, we examined the miscibility of DA and various imidazolium-based ionic liquids such as 1-ethyl-3-methylimidazolium bis(trifluoromethylsulfonyl)imide ([EMIM][TFSI]) and 1-butyl-3-methylimidazolium tetrafluoroborate ([BMIM][BF₄]). The combination of DA and 1-ethyl-3-methylimidazolium tetrafluoroborate ([EMIM][BF₄]) (IL) in the various compositions provided the room-temperature columnar liquid crystals, whereas other ionic liquid mixtures became liquids at ambient temperature. The size, shape, and ionicity of ionic

liquids are delicate for the induction of the columnar liquid-crystalline phase. IL is one of the promising liquid electrolytes for fabricating nanostructured actuators because it exhibits the highest ionic conductivity among the conventional ionic liquids.^{13,27–29}

Figure 2 shows the phase diagram of the binary mixtures of DA and IL (DA/IL(X)) as a function of the mole % (X) of IL

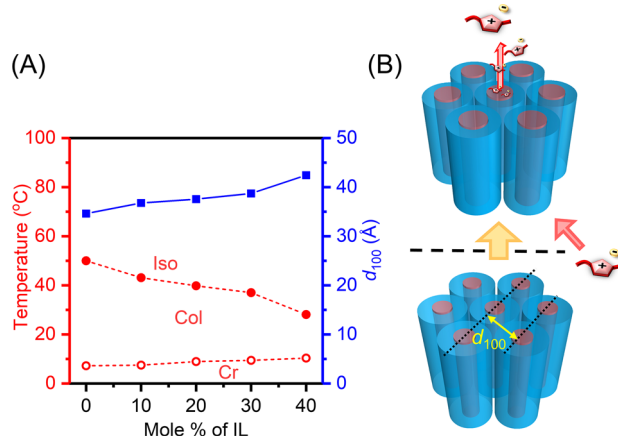


Figure 2. (A) Phase-transition temperatures of the binary mixtures of DA and IL as a function of the mole % of IL and the d -spacing value assigned to the X-ray diffraction of the (100) plane of the columnar phase. Iso, Col, and Cr are isotropic liquid, columnar, and crystalline, respectively. (B) Illustration of the swelling in nonpolymerized columnar ionic channels by incorporating ionic liquid.

in the mixture and the plots of the d -spacing value assigned to the X-ray diffraction from the (100) plane of the columnar (Col) phase. Homogeneous mixtures forming the Col phase are obtained for up to 40 mol % of IL. As the IL content increases, the temperature range of the Col phase decreases. For example, DA/IL(30) shows the Col-Iso phase transition at 36.9 °C (Figure S1), whereas the transition of DA alone is observed at 50.0 °C (Table S1). In addition, the intercolumnar distance is increased by the addition of IL. Monomeric DA gives a d_{100} value of 36.8 Å, and the d_{100} value for monomeric DA/IL(40) is 42.4 Å. These results suggest that the incorporation of IL in the center of the columns results in the formation of highly mobile 1D ion channels with larger diameter.

The monomeric mixture DA/IL(X) (X = 10–40) formed the polydomain columnar structures in a sandwiched glass cell with a gap of 35 μ m by slow cooling from the isotropic liquid state, although we previously achieved the homeotropic alignment of columns up to 10 μ m thickness for DA alone. The polarized optical microscopic image of DA/IL(30) (Figure S2A) shows a mixed texture with black regions indicating vertical alignment and birefringent regions due to planar and tilted columnar orientations.

The photopolymerization kinetics of DA/IL(X) in the Col phases were examined to obtain highly ion-conductive polymer films with sufficient mechanical properties to be used as iEAP actuators. Prolonged UV irradiation at a power density of 5.5 mW/cm² for 10 min resulted in a gradual decrease in the ionic conductivity due to photopolymerization (Figure S3). A free-standing and flexible polymer film preserving the Col orientation (Figure S2B) was successfully obtained by UV irradiation for 150 s. The Fourier transform infrared (FT-IR)

spectrum confirmed the polymerization, as the stretching bands of acrylate at 1636 and 811 cm^{-1} were hardly observed (Figure S4). In addition, the photopolymerization of monomeric mixtures in the Iso phases produced flexible amorphous films. Hereafter, the polymer films obtained by photo-cross-linking DA/IL(X) in the Col and Iso phases are denoted as $\text{P}_{\text{Col}}/\text{IL}(\text{X})$ and $\text{P}_{\text{Iso}}/\text{IL}(\text{X})$, respectively. The ionic conductivity for the polymer films increased with an increase in the mole % of IL (Figure 3A). The activation energies (E_a) for

transport. Moreover, $\text{P}_{\text{Col}}/\text{IL}(30)$ maintains the anisotropic structure up to 150 $^{\circ}\text{C}$ and possesses a glass-transition temperature at -14 $^{\circ}\text{C}$ that is attributed to the assembled ionic moieties (Figure S5). Alignment of the columns is essential to achieve high ionic conductivity. The columnar nanostructured polymer film with a columnar orientation parallel to the film surface ($\text{P}_{\text{Col}}/\text{IL}(30)$), which was obtained by mechanical shearing of DA/IL(30) and the subsequent photopolymerization, exhibited six times lower ionic conductivity ($1.06 \times 10^{-7} \text{ S cm}^{-1}$) across the membrane than the $\text{P}_{\text{Col}}/\text{IL}(30)$ film with randomly oriented columns (Figure S6). The concept of nanosegregation, consisting of liquid-like, ion-conductive, and rigid mesogenic domains, may lead to new designs of iEAP actuators that combine high elasticity for power generation and high conductivity for bendability.

Here we show the enhanced bending strain and blocking force for the iEAP actuator using the photo-cross-linked ionic columnar liquid-crystalline polymer film compared with those of the amorphous film actuator (Figure 4). The polymer films containing 30 mol % of IL were examined because of the more stable confinement of IL in the membrane. On the contrary, partial leakage of IL was observed for the polymer films containing 40 mol % of IL during the peeling process from the glass substrate. The ion-conductive polymer films (thickness 35 μm) were sandwiched between a pair of flexible PEDOT:PSS film electrodes with a thickness of $7 \pm 2 \mu\text{m}$ and hand-pressed.

Figure 4A shows the bending displacement of the actuator strip under the application of an alternating current (AC) square voltage of ± 2 V at a frequency of 0.01 Hz under ambient conditions. The actuator strip showed a bending motion to the anode side. The size of the imidazolium cation of IL is larger than that of the tetrafluoroborate anion. It has been reported that the cationic transference number in neat IL is larger than that of the anion by pulsed gradient spin echo-nuclear magnetic resonance (PGSE-NMR) experiments.^{30,31}

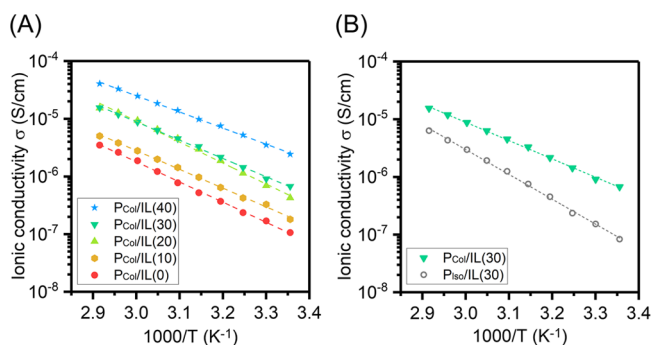


Figure 3. (A) Temperature dependence of ionic conductivity for the columnar nanostructured polymer films of $\text{P}_{\text{Col}}/\text{IL}(\text{X})$ with different mole % of IL. (B) Comparison of the conductivities for the columnar polymer film $\text{P}_{\text{Col}}/\text{IL}(30)$ (triangle) and the amorphous polymer film $\text{P}_{\text{Iso}}/\text{IL}(30)$ (open circle). The dashed lines are fit by the Arrhenius equation.

$\text{P}_{\text{Col}}/\text{IL}(0)$ and $\text{P}_{\text{Col}}/\text{IL}(30)$ are estimated to be 29 and 26 kJ/mol, respectively. It is noteworthy that $\text{P}_{\text{Col}}/\text{IL}(30)$ exhibits eight times higher ionic conductivity at 25 $^{\circ}\text{C}$ compared with $\text{P}_{\text{Iso}}/\text{IL}(30)$ (Figure 3B). In addition, the activation energy of $\text{P}_{\text{Col}}/\text{IL}(30)$ is smaller than that of $\text{P}_{\text{Iso}}/\text{IL}(30)$ ($E_a = 36$ kJ/mol). The formation of columnar ionic channels confining ionic liquids is a significant key to achieving efficient ion

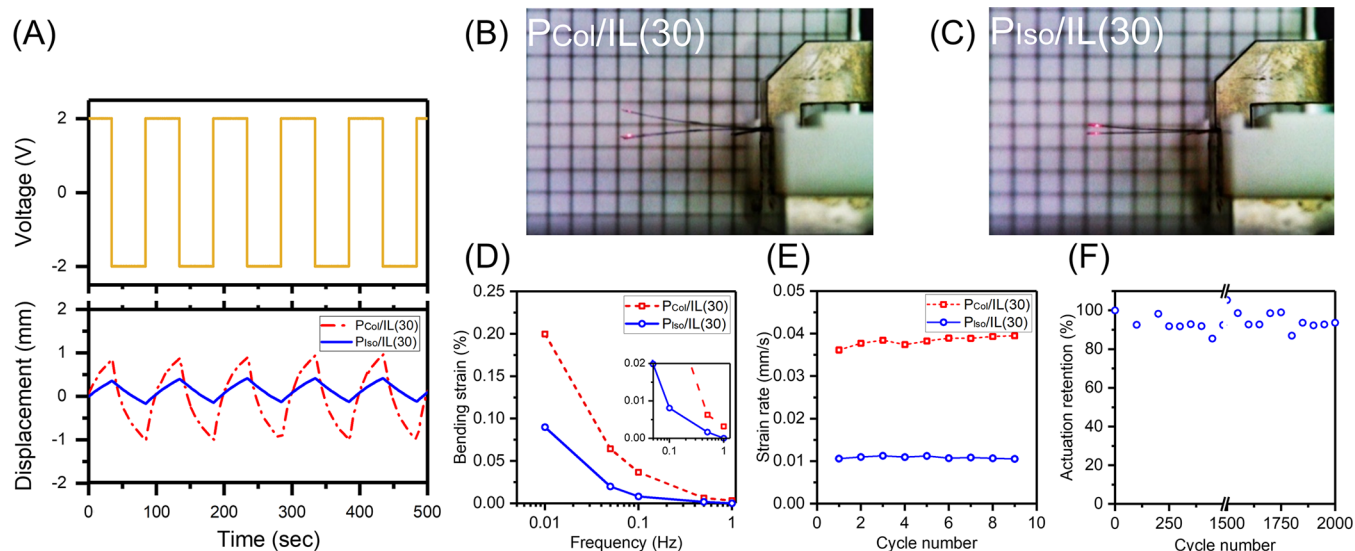


Figure 4. Actuation response of the columnar and amorphous polymer film electrolytes sandwiched between PEDOT:PSS electrodes. (A) Applied voltage and tip displacement under an AC voltage of 2 V at a frequency of 0.01 Hz. (B,C) Photographs of the columnar and amorphous actuators bending at ± 2 V and 0.01 Hz, respectively. The grid size is 2 mm. (D) Bending strain (%) of actuators at AC 2 V with the frequency range from 0.01 to 1 Hz. (E) Strain rate of the actuators derived from the displacement versus time plots in panel A (bottom). (F) Cyclic stability of the columnar polymer film actuator under 2 V at 1 Hz. The bending strain for each cycle is normalized by the initial strain value.

Therefore, the faster migration of cations confined in the channels to the cathode may cause the swelling of the cathode layer, although the polarization of the columnar liquid-crystalline polymer under electric fields should also contribute to the bending displacement. In fact, the trilayered actuator strip of $P_{\text{Col}}/\text{IL}(0)$ without ionic liquid and PEDOT:PSS showed a slight bending actuation upon the application of voltages (maximum displacement of 30 μm under 2 V at 0.01 Hz). To gain further insight into the actuation mechanism, we conducted cyclic voltammetry measurements for the $P_{\text{Col}}/\text{IL}(0)$ and $P_{\text{Col}}/\text{IL}(30)$ -based actuators. The cyclic voltammogram of the $P_{\text{Col}}/\text{IL}(30)$ actuator shows a capacitive current–voltage behavior (Figure S7), although the capacitive saturation of the current is not reached, as resistive contributions decrease the overall current. In addition, no Faradaic current due to the redox reactions between IL and PEDOT:PSS is seen. On the contrary, for the $P_{\text{Col}}/\text{IL}(0)$ without ionic liquids, no capacitive or Faradaic currents are observed. These results indicate that the columnar liquid-crystalline polymer matrix ($P_{\text{Col}}/\text{IL}(0)$) served only as a nanostructured reservoir for the ionic liquid.

The maximum peak-to-peak tip displacement of $P_{\text{Col}}/\text{IL}(30)$ -based actuator reaches 2 mm, whereas the $P_{\text{Iso}}/\text{IL}(30)$ -based actuator shows a maximum displacement of 0.5 mm. In addition, the $P_{\text{Col}}(\text{//})/\text{IL}(30)$ actuator with a parallel orientation of columns to the film surface shows only a 0.2 mm displacement under the same conditions (Figure S8). The orientation of the ionic channel significantly affected the bending behaviors. Figure 4D shows the peak-to-peak bending strain of the actuators based on $P_{\text{Col}}/\text{IL}(30)$ and $P_{\text{Iso}}/\text{IL}(30)$ under the application of 2 V with a frequency of 0.01, 0.05, 0.1, 0.5, and 1 Hz. The bending strain was calculated by the following equation: $\epsilon = 2\delta d \times 100/(\delta^2 + L^2)$,³² where δ is the peak-to-peak displacement, d is the total thickness of actuator, and L is the free length of the strip. At a frequency of 0.01 Hz, the bending strain of the $P_{\text{Col}}/\text{IL}(30)$ -based actuator ($\epsilon = 0.20\%$) was about two times larger than that of the $P_{\text{Iso}}/\text{IL}(30)$ -based actuator ($\epsilon = 0.09\%$). This may be due to the increased ion migration through the partially continuous ion channels from the nanostructured polymer electrolyte to the electrode surface. For $P_{\text{Iso}}/\text{IL}(30)$, the ionic liquid may be encapsulated in insulating and rigid polymerized mesogenic domains, which leads to the very slow diffusion of ions. In contrast, the $P_{\text{Col}}/\text{IL}(30)$ -based actuator at 1 Hz exhibited a bending motion with a strain value of 0.0032%, whereas no bending behavior was observed for $P_{\text{Iso}}/\text{IL}(30)$. These results indicate that at frequencies higher than 1 Hz, the disordered polymer matrix cannot store enough charges in the electrodes to cause bending actuation, even though it contains the same number of ions as the columnar polymer matrix. The frequency dependence of the curvature for bending actuators also shows the same trend as the bending strain (Figure S9). In addition, the $P_{\text{Col}}/\text{IL}(30)$ -based columnar actuator showed a 3.5 times faster response than the amorphous $P_{\text{Iso}}/\text{IL}(30)$ -based actuator (Figure 4E). The $P_{\text{Col}}/\text{IL}(30)$ -based columnar actuator also exhibited fair cycling stability under 2 V at 1 Hz with strain degradation of 8% after 2000 cycles (Figure 4F).

To further investigate the relationship between the electrolyte morphologies and the actuation response, we measured the time-dependent change of deformation under direct current (DC) voltages of 1 and 2 V (Figure S10). With the increase in the applied voltages, the maximum displacements were increased probably due to the increase in the capacitive

charges. For the amorphous actuator with a free length of 1 cm, the displacement gradually increased over 800 s and reached a steady-state value of 0.62 mm at 1 V and 0.8 mm at 2 V. In contrast, the columnar actuator exhibited a larger maximum displacement in a shorter time than the amorphous actuator (1.14 mm for 420 s at 1 V and 1.8 mm for 150 s at 2 V). As for the columnar actuator at 2 V, a small back relaxation³³ toward the negative electrode over 150 s was observed.

The $P_{\text{Col}}/\text{IL}(30)$ -based columnar actuator can yield a blocking force of 0.28 mN when an AC bias of ± 2 V at 0.01 Hz is applied (Figure 5A), which is two times larger than that

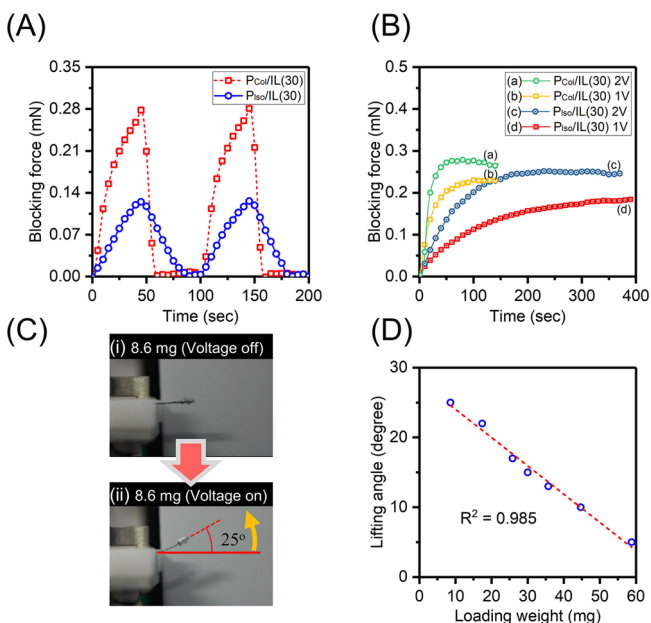


Figure 5. Blocking force for the columnar and amorphous polymer actuators under (A) AC voltage of 2 V at 0.01 Hz and (B) DC voltage of 1 and 2 V. (C) Photograph of the columnar polymer actuator lifting 8.6 mg of aluminum foil under DC voltage of 2 V. The dimension of the actuator strip is 12 mm $\times$ 4 mm $\times$ 61 μm . The weight of the actuator is 2.9 mg. (D) Lifting angle of the columnar actuator versus loaded weight.

obtained for the $P_{\text{Iso}}/\text{IL}(30)$ -based actuator. The generated force was reduced by decreasing the applied voltage (Figure 5B and Table S2) and controlled with step changes in voltage (Figure S11). Moreover, the lifting of an aluminum foil of 10–60 mg was successfully demonstrated for the $P_{\text{Col}}/\text{IL}(30)$ -based columnar actuator (size: 12 mm $\times$ 4 mm $\times$ 61 μm , weight: 2.9 mg) under the application of DC 2 V (Figure 5C and Figure S12), and the actuator was able to lift objects at least three times its own weight. The lifting angle of the actuator was proportional to the load weight (Figure 5D). The force generation of the $P_{\text{Col}}/\text{IL}(30)$ -based columnar actuator containing 7.9 wt % of IL is comparable to that of polymer ion gel actuators, composed of thermoplastic polyurethane containing 80 wt % of ionic liquid (1-ethyl-3-methylimidazolium bis(trifluoromethylsulfonyl)imide) and PEDOT:PSS electrode (0.32 mN at 2 V)³⁴ and zwitterion-doped, single-cation-conducting block copolymers sandwiched by single-walled carbon nanotube electrodes (0.3 mN at 1 V).¹⁰ It is noteworthy that only a small number of ions confined in the nanoscale space can generate the bending motion and force,

which can contribute to reducing the risk of ionic liquid leakage and the production cost.

In summary, we have reported photo-cross-linked ionic columnar liquid-crystalline polymer films confining ionic liquid into 1D channels and the first application to electroactive actuators that shows low-voltage-driven bendability in the air and yields controlled force generation. We have unveiled a direct correlation between the nanochannels and the actuation performances. The concept of nanosegregation, consisting of liquid-like, ion-conductive, and rigid mesogenic domains, may lead to new designs for ionic electroactive actuators to achieve the conflicting functions of high-power generation and fast motion.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acsmaterialslett.1c00565>.

Preparation of liquid-crystalline mixtures, photopolymerized films, PEDOT:PSS films, and actuator films, details of measurements, POM images, differential scanning calorimetry (DSC) thermograms, FT-IR spectra, ionic conductivities, cyclic voltammograms, and actuation performance (PDF)

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Notes

The authors declare no competing financial interest.

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■ REFERENCES

- (1) Asaka, K.; Okuzaki, H. *Soft Actuators: Materials, Modeling, Applications, and Future Perspectives*, 2nd ed.; Springer: Singapore, 2019.
- (2) Feng, C.; Hemantha Rajapaksha, C. P.; Jákli, A. Ionic Elastomers for Electric Actuators and Sensors. *Engineering* **2021**, *7*, 581–602.
- (3) Kim, O.; Kim, S. J.; Park, M. J. Low-Voltage-Driven Soft Actuators. *Chem. Commun.* **2018**, *54*, 4895–4904.
- (4) White, B. T.; Long, T. E. Advanced in Polymeric Materials for Electromechanical Devices. *Macromol. Rapid Commun.* **2019**, *40*, 1800521.
- (5) Martins, P.; Correia, D. M.; Correia, V.; Lanceros-Mendez, S. Polymer-Based Actuators: Back to the Future. *Phys. Chem. Chem. Phys.* **2020**, *22*, 15163–15182.
- (6) Ueki, T.; Watanabe, M. Macromolecules in Ionic Liquids: Progress, Challenges, and Opportunities. *Macromolecules* **2008**, *41*, 3739–3749.
- (7) Fukushima, T.; Asaka, K.; Kosaka, A.; Aida, T. Fully Plastic Actuator through Layer-by-Layer Casting with Ionic-Liquid-Based Bucky Gel. *Angew. Chem., Int. Ed.* **2005**, *44*, 2410–2413.
- (8) Duncan, A. J.; Leo, D. J.; Long, T. E. Beyond Nafion: Charged Macromolecules Tailored for Performance as Ionic Polymer Transducers. *Macromolecules* **2008**, *41*, 7765–7775.
- (9) Kim, O.; Shin, T. J.; Park, M. J. Fast Low-Voltage Electroactive Actuators Using Nanostructured Polymer Electrolytes. *Nat. Commun.* **2013**, *4*, 2208.
- (10) Kim, O.; Kim, H.; Choi, U. H.; Park, M. J. One-Volt-Driven Superfast Polymer Actuators Based on Single-Ion Conductors. *Nat. Commun.* **2016**, *7*, 13576.
- (11) Lee, J.-W.; Yu, S.; Hong, S. M.; Koo, C. M. High-Strain Air-Working Soft Transducers Produced from Nanostructured Block Copolymerionomer/Silicate/Ionic Liquid Nanocomposite Membranes. *J. Mater. Chem. C* **2013**, *1*, 3784–3793.
- (12) Imaizumi, S.; Kokubo, H.; Watanabe, M. Polymer Actuators Using Ion-Gel Electrolytes Prepared by Self-Assembly of ABA-Triblock Copolymers. *Macromolecules* **2012**, *45*, 401–409.
- (13) Guo, D.; Han, Y.; Huang, J.; Meng, E.; Ma, L.; Zhang, H.; Ding, Y. Hydrophilic Poly(vinylidene Fluoride) Film with Enhanced Inner Channels for Both Water- and Ionic Liquid-Driven Ion-Exchange Polymer Metal Composite Actuators. *ACS Appl. Mater. Interfaces* **2019**, *11*, 2386–2397.
- (14) Simaite, A.; Tondou, B.; Souères, P.; Bergaud, C. Hybrid PVDF/PVDF-graft-PEGMA Membranes for Improved Interface Strength and Lifetime of PEDOT:PSS/PVDF/Ionic Liquid Actuators. *ACS Appl. Mater. Interfaces* **2015**, *7*, 19966–19977.
- (15) Kim, J.; Jeon, J.-H.; Kim, H.-J.; Lim, H.; Oh, I.-K. Durable and Water-Floatable Ionic Polymer Actuator with Hydrophobic and Asymmetrically Laser-Scribed Reduced Graphene Oxide Paper Electrodes. *ACS Nano* **2014**, *8*, 2986–2997.
- (16) Kato, T.; Yoshio, M.; Ichikawa, T.; Soberats, B.; Ohno, H.; Funahashi, M. Transport of Ions and Electrons in Nanostructured Liquid Crystals. *Nat. Rev. Mater.* **2017**, *2*, 17001.
- (17) Cho, B.-K. Nanostructured Organic Electrolytes. *RSC Adv.* **2014**, *4*, 395–405.
- (18) Kerr, R. L.; Miller, S. A.; Shoemaker, R. K.; Elliott, B. J.; Gin, D. L. New Type of Li Ion Conductor with 3D Interconnected Nanopores via Polymerization of a Liquid Organic Electrolyte-Filled Lyotropic Liquid-Crystal Assembly. *J. Am. Chem. Soc.* **2009**, *131*, 15972–15973.
- (19) Van Kuringen, H. P. C.; Eikelboom, G. M.; Shishmanova, I. K.; Broer, D. J.; Schenning, A. P. H. J. Responsive Nanoporous Smectic Liquid Crystal Polymer Networks as Efficient and Selective Adsorbents. *Adv. Funct. Mater.* **2014**, *24*, 5045–5051.
- (20) Wang, S.; Liu, X.; Wang, A.; Wang, Z.; Chen, J.; Zeng, Q.; Wang, X.; Zhang, L. An Ionic Liquid Crystal-Based Solid Polymer Electrolyte with Desirable Ion-Conducting Channels for Superior Performance Ambient-Temperature Lithium Batteries. *Polym. Chem.* **2018**, *9*, 4674–4682.
- (21) Feng, C.; Rajapaksha, C. P. H.; Cedillo, J. M.; Piedrahita, C.; Cao, J.; Kaphle, V.; Lüssem, B.; Kyu, T.; Jákli, A. Electroresponsive Ionic Liquid Crystal Elastomers. *Macromol. Rapid Commun.* **2019**, *40*, 1900299.
- (22) Yoshio, M.; Kagata, T.; Hoshino, K.; Mukai, T.; Ohno, H.; Kato, T. One-Dimensional Ion-Conductive Polymer Films: Alignment and Fixation of Ionic Channels Formed by Self-Organization of Polymerizable Columnar Liquid Crystals. *J. Am. Chem. Soc.* **2006**, *128*, 5570–5577.

- (23) Ichikawa, T.; Yoshio, M.; Hamasaki, A.; Kagimoto, J.; Ohno, H.; Kato, T. 3D Interconnected Ionic Nano-Channels Formed in Polymer Films: Self-Organization and Polymerization of Thermotropic Bicontinuous Cubic Liquid Crystals. *J. Am. Chem. Soc.* **2011**, *133*, 2163–2169.
- (24) Gin, D. L.; Bara, J. E.; Noble, R. D.; Elliott, B. J. Polymerized Lyotropic Liquid Crystal Assemblies for Membrane Applications. *Macromol. Rapid Commun.* **2008**, *29*, 367–389.
- (25) Lugger, J.; Mulder, D. J.; Sijbesma, R.; Schenning, A. Nanoporous Polymers Based on Liquid Crystals. *Materials* **2018**, *11*, 104.
- (26) Sakamoto, T.; Ogawa, T.; Nada, H.; Nakatsuji, K.; Mitani, M.; Soberats, B.; Kawata, K.; Yoshio, M.; Tomioka, H.; Sasaki, T.; Kimura, M.; Henmi, M.; Kato, T. Development of Nanostructured Water Treatment Membranes Based on Thermotropic Liquid Crystals: Molecular Design of Sub-Nanoporous Materials. *Adv. Sci.* **2018**, *5*, 1700405.
- (27) Takeuchi, I.; Asaka, K.; Kiyohara, K.; Sugino, T.; Terasawa, N.; Mukai, K.; Fukushima, T.; Aida, T. Electromechanical Behavior of Fully Plastic Actuators based on Bucky Gel Containing Various Internal Ionic Liquids. *Electrochim. Acta* **2009**, *54*, 1762–1768.
- (28) Ue, M.; Murakami, A.; Nakamura, S. A Convenient Method to Estimate Ion Size for Electrolyte Materials Design. *J. Electrochem. Soc.* **2002**, *149*, A1385–A1388.
- (29) Hagiwara, R.; Lee, J. S. Ionic Liquids for Electrochemical Devices. *Electrochemistry* **2007**, *75*, 23–34.
- (30) Noda, A.; Hayamizu, K.; Watanabe, M. Pulsed-Gradient Spin–Echo ^1H and ^{19}F NMR Ionic Diffusion Coefficient, Viscosity, and Ionic Conductivity of Non-Chloroaluminate Room-Temperature Ionic Liquids. *J. Phys. Chem. B* **2001**, *105*, 4603–4610.
- (31) Hayamizu, K.; Aihara, Y.; Nakagawa, H.; Nukuda, T.; Price, W. S. Ionic Conduction and Ion Diffusion in Binary Room-Temperature Ionic Liquids Composed of [emim][BF₄] and LiBF₄. *J. Phys. Chem. B* **2004**, *108*, 19527–19532.
- (32) Mukai, K.; Asaka, K.; Sugino, T.; Kiyohara, K.; Takeuchi, I.; Terasawa, N.; Futaba, D. N.; Hata, K.; Fukushima, T.; Aida, T. Highly Conductive Sheets from Millimeter-Long Single-Walled Carbon Nanotubes and Ionic Liquids: Application to Fast-Moving, Low-Voltage Electromechanical Actuators Operable in Air. *Adv. Mater.* **2009**, *21*, 1582–1585.
- (33) Nemat-Nasser, S.; Wu, Y. Comparative Experimental Study of Ionic Polymer-Metal Composites with Different Backbone Ionomers and in Various Cation Forms. *J. Appl. Phys.* **2003**, *93*, S255–S267.
- (34) Kim, S. Y.; Kim, Y.; Cho, C.; Choi, H.; Park, H. W.; Lee, D.; Heo, E.; Park, S.; Lee, H.; Kim, D. H. Deformable Ionic Polymer Artificial Mechanotransducer with an Interpenetrating Nanofibrillar Network. *ACS Appl. Mater. Interfaces* **2019**, *11*, 29350–29359.